

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Seat No.

Midterm Examination of Semester 1

Year: 2020

Subject: 392131 Physics I

Section: 15 - 18

Date: 31 August 2020

Time: 13.00 - 15.00

Name: _____ ID: _____ Class: _____

Directions: The test is designed to measure your comprehension. The test has 1 section. There will be 6 pages (including this page) and they are worth 40 points.

- This exam paper contains no errors. If a suspected error is found, it is the student's discretion to correct it.
- Answer the questions on this test paper.
- Books, documents and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators can be used in this test.

Now begin the test.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

1. Perform the following unit conversions

(2 points)

a) $52.43 \text{ MB} = \underline{\hspace{2cm}}$ TB (Note: "B" = byte)

b) $1 \text{ kg}\cdot\text{m}/\text{min} = \underline{\hspace{2cm}}$ $\text{g}\cdot\text{cm}/\text{s}$

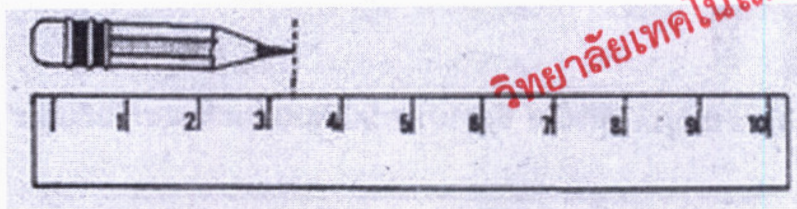
2. A car is traveling at a speed of 40.0 m/s on an interstate highway where the speed limit is 75.0 mi/h . Is the driver exceeding the speed limit? Justify your answer.

($1 \text{ mi} = 1.6 \text{ km}$)

(2 points)

3. What is the length of the pencil?

(1 point)



4. How many significant figures are there in the follow questions?

(2 points)

a) 0.08945

b) 18000.00

c) 89.90 ± 0.02

d) 5.41203×10^9

5. Estimate the value of the radius of Hydrogen atom $5.0 \times 10^{-11} \text{ m}$

(1 points)

6. Use the rules for significant figures to find the answer to the problem (2 points)

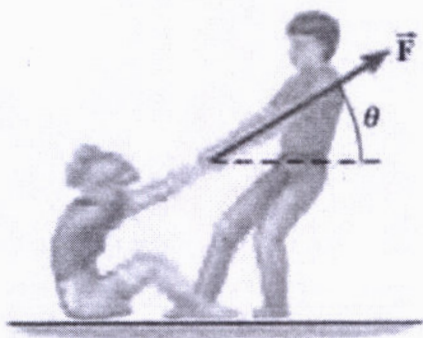
$0.055 + 17.0 + 111 - 3.301 =$

7. Use the rules for significant figures to find the answer of this problem (2 points)

$$(456 \times 11.1) \div (2.0) =$$

8. A rectangular piece of aluminum is 7.50 ± 0.01 cm long and 1.50 ± 0.01 cm wide. Find the area of the rectangle and the uncertainty in the area. (3 points)

9. Your friend has slipped and fallen. To help her up, you pull with a force $\vec{F}$, as the drawing shows. The vertical component of this force is 400 N, and the horizontal component is 300 N, Find (3 points)



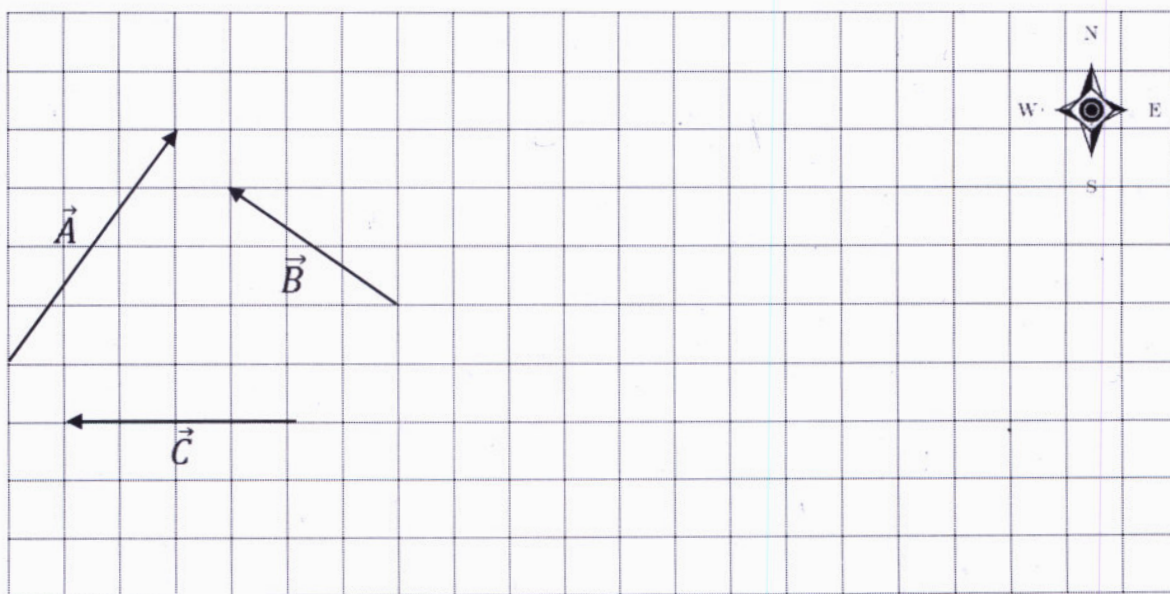
a) the magnitude of F

b) the angle (θ)

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10. Determine the resultant vector of $\vec{A} - \vec{B} - \vec{C}$

(2 points)



11. A car moves at a constant speed of 10 m/s for 10 seconds and then stepped on the brake, causing it to stop at the junction in 5 seconds. (4 points)

a) Drawing graph displacement with time, graph velocity with time and graph acceleration with time.

b) What is average velocity in this trip?

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12. A stone falls from a balloon that a uniform speed upward of 3 m/s at high 50 m from ground. What is highest distance of the stone from the ground? (2 points)

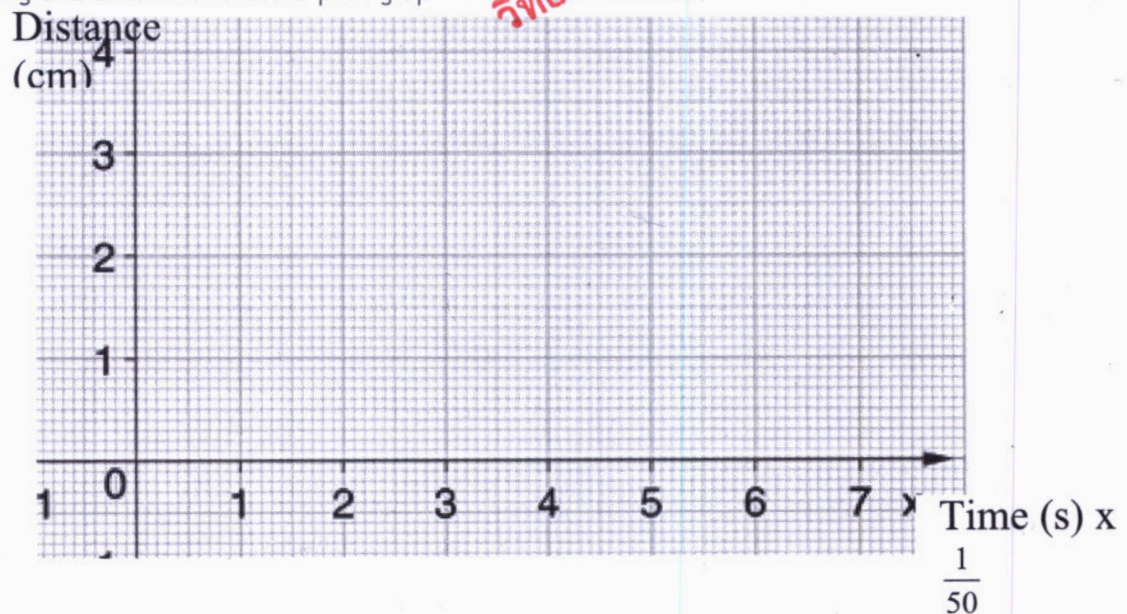
13. In the test of the movement of experimental car attached to a paper via a time-trigger with a frequency of 50 Hz, as shown in the picture. (4 points)



The distance on each points of paper measured from point A is shown in the table in below.

Point	distance from point A (cm)
A	0.0
B	0.1
C	0.5
D	1.1
E	1.8
F	2.5
G	3.3

- a) Using the data in the table plot graph distance with time



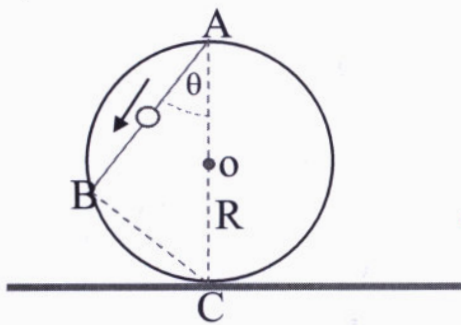
- b) Find velocity at Point E from data at table and from graph.

13. A ball thrown straight up takes 2.0 s to reach a height of 40 m. Find (a) its initial speed, (b) its speed at this height, and (c) how much higher the ball will go.

Take $g = 10 \text{ m/s}^2$.

(5 points)

15. A frictionless wire AB is fixed on a sphere of radius R . A very small spherical ball slips on this wire. What time taken by this ball to slip from A to B? (5 points)



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. Good Luck

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